



Ankur Gupta
Electrical Engineering
Indian Institute of Technology, Bombay
Specialization: Integrated Circuit and Systems (EE6)

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Examination	University	Institute	Year	CPI/%
Ph.D.	IIT Bombay	IIT BOMBAY	2023–2027	8.65
Post Graduation	NIT Rourkela	NIT Rourkela	2020–2022	8.42
Graduation	RGPV Bhopal	ITM Gwalior	2014–2018	7.04
Intermediate/+2	UP Board	SVM Inter College	2011–13	76.2

AREAS OF INTEREST

- Digital VLSI Design
- Processor Design
- FPGA Based Acceleration
- Physical Design
- DFT

SCHOLASTIC ACHIEVEMENTS

- Published a paper titled **A Comparative Study of Adaptive Channel Selection for Resource-Constrained On-Chip Seizure Detection** at the *IEEE Asia Pacific Conference on Circuits and Systems*, Busan, South Korea, 2025.
- Manuscript under review: **A 16-Channel Rapid On-Chip Seizure Classifier with Online Re-Training for Closed-Loop Neuromodulation**, submitted to *IEEE International Symposium on Circuits and Systems (ISCAS) 2026*.
- Secured **First Runner-Up** in the **Smart India Hackathon 2017** by developing a **Wheelchair control system using EOG signals for individuals with disabilities**.
- Secured **First Prize** in a **PID-Based Line Follower Competition** organized by ITM Vadodara.
- Secured an All India Rank 3492 in the **Graduate Aptitude Test in Engineering (GATE) 2020**.
- Contributed bug fixes to the open-source **facebookresearch/Fairchem GitHub** repository.
- Added automation scripts in **FOSSEE/eSim GitHub** repository to enable seamless installation of open-source PDKs via Volare.

RESEARCH WORK

- **On-Chip Resource Efficient Classifiers for Neural Application** | *PhD's Thesis*
Guide : Prof. Laxmeesha Somappa, Electrical Engineering, IIT Bombay (Jan'23 - present)
- **Completed Work:**
 - Designed and architected a **16-channel logistic regression–based on-chip patient specific Seizure Detector** with an **online retraining unit** to enable adaptive closed-loop neuromodulation.
 - Developed a **shared multiply–accumulate (MAC) architecture** to reduce area and power while maintaining real-time throughput.
 - Completed the **full RTL-to-GDSII flow**, including RTL design, functional verification, synthesis, place-and-route, and sign-off.
- **Ongoing and Future Work:**
 - Designing and training a **lightweight CNN architecture** for **patient-independent Seizure Detection**.
 - Exploring and modeling different **hardware accelerator architectures** for **lightweight CNN models** using **SystemC**.
 - Will develop a **scalable CNN accelerator** capable of processing **arbitrary CNN layers** by integrating a **PicoRV32 RISC-V** processor for flexible control and **functionally validating** the PE modules on a **FPGA**.
 - Architecting the PicoRV32-based control subsystem to manage the **accelerator datapath** and **memory system**, defining the **HW/SW interface** and **custom instruction set** for end-to-end CNN execution.
 - Will execute the complete **RTL-to-GDSII** physical design flow for **TSMC 65nm** node, including **synthesis**, **Place & Route (P&R)**, and **Static Timing Analysis (STA)** to prepare the **ASIC** for tape-out.
- **Implementation of DFT in Quantum Computer** | *M.Tech*
Guide : Prof. Supratim Gupta, Electrical Engineering, NIT Rourkela (July'21 - June'22)
 - Developed an **FPGA-based emulator for quantum computing**, enabling the emulation of **quantum algorithms**, and applied **Discrete Fourier Transform (DFT)-based methods** for quantum computation.

KEY ACADEMIC PROJECTS & ASSIGNMENTS

- **MAC unit with SPI** | *Course Project*
Course: *Hardware Description* | Prof. Sachin Patkar (Nov'23)

- Designed a **SPI Protocol** for MAC operation using MAC unit.
- MAC unit designed using Dadda multiplier with Brent-Kung adder.
- Utilized **Verilog HDL** for design using **Vivado** and conducted verification with a specific testbench.
- **Pattern Detection using Neural-Network on FPGA Pynq-z2** | *Course Project* (Apr'23)
Course: *Embedded Systems Design* | *Prof. Laxmeesha Somappa*
 - Designed a **neural network accelerator** interfaced through **AXI-Stream** and controlled by a **Zynq processor**.

RELEVANT COURSES TAKEN

- VLSI Design
- Hardware Description
- Testing and Verification
- Embedded System Design
- VLSI Design Lab
- Introduction to Machine Learning

WORK EXPERIENCE

- **Associate Engineer** | AFFINE (Aug'22 - Dec'23)
 - Contributed in **Data Visualization on Map** project using Google cloud VM with login system using USER-ID and OTP.

TECHNICAL SKILLS

- **Tools:** Altera Quartus Prime, Xilinx Vivado, ModelSim, OpenLane, Yosys, Cadence Xcelium, Genus, Innovus, Virtuoso, Ngspice, LTspice, Icarus Verilog, GTKWave, CMU-BDD, Linux OS, Yosys, MiniSAT, MATLAB, Keil, Proteus, Xschem, Magic, $\text{\LaTeX}$.
- **Hardware Platforms:** PYNQ-Z2 Board, Nexys A7, Raspberry Pi, Atmega328p Micro-controller, TM4GH123 Micro-controller, ESP32 Micro-controller
- **Programming Languages:** Verilog, VHDL, C, C++, Python, TCL, BASH, SystemC, Makefile.

POSITIONS OF RESPONSIBILITY

- **Institute Teaching Assistant** | Digital Circuits Lab, IIT Bombay (Aug'23 - present)
 - Assisted students in writing and synthesizing **VHDL code**, and in testing designs on **FPGA** using **JTAG**, with verification against **golden outputs**.
- **Institute Teaching Assistant** | Embedded Systems Design, IIT Bombay (Jan'24 - present)
 - Guided **Embedded C** and **VHDL/Verilog**, taught **Code Composer Studio**, **Vivado** and **Vitis SDK**, and designed practical assignments for evaluation.
- **Institute Teaching Assistant** | VLSI Design, IIT Bombay (Aug'24 - present)
 - Delivered hands-on tutorials on **Magic**, **ngspice**, **OpenLane**, and **Cadence** using the **SKY130/TSMC65 PDK**, and developed practical assignments for students.
- **Institute Teaching Assistant** | VLSI Design LAB, IIT Bombay (Jan'25 - present)
 - Conducted hands-on tutorials on **Vivado** covering **ILA**, **VIO**, and **BRAM**; guided students in **RISC-V processor** implementation on **FPGA**; and demonstrated the **RTL-to-GDS flow** using **Cadence** and **OpenLane**, followed by the design of practical assignments.
- **Institute Interview Coordinator** | IIT Bombay (Nov'25 - Dec'25)

EXTRACURRICULAR ACTIVITY

- **Hobbies:** Swimming, Books Reading, Rubik-cube solving, Bonsai, Trekking, Watching Movies.